

Lignans and isoflavonoids in plasma and prostatic fluid in men: samples from Portugal, Hong Kong, and the United Kingdom.

BACKGROUND: Chinese men have lower incidences of prostate cancer compared to men from Europe and North America. Asians consume large quantities of soya, a rich source of isoflavonoids phyto-oestrogens and have high plasma and urinary levels of these compounds. The mammalian lignans, enterolactone and enterodiol, are another group of weak plant oestrogens and are derived from seeds, cereals and grains. Vegetarians have high plasma and urinary concentrations of lignans. **METHODS:** The concentrations lignans and isoflavonic phyto-oestrogens were determined by gas chromatography-mass spectrometry (GC-MS) in plasma and prostatic fluid from Portuguese, Chinese and British men consuming their traditional diets. **RESULTS:** In prostatic fluid the mean concentrations of enterolactone were 31, 162 and 20.3 ng/ml for Hong Kong, Portugal and Britain respectively. Very high levels of enterolactone (> 600 ng/ml) were observed in the prostatic fluid of some of the men from Portugal. High concentrations of equol (3270 ng/ml) and daidzein (532 ng/ml) were found in a sample of prostatic fluid from Hong Kong. Higher mean levels of daidzein were observed in prostatic fluid from Hong Kong at 70 ng/ml, compared to 4.6 and 11.3 ng/ml in samples from Portugal and Britain respectively. Mean levels of daidzein were higher in the plasma samples from Hong Kong (31.3 ng/ml) compared to those from Portugal (1.3 ng/ml) and Britain (8.2 ng/ml). In general, the mean plasma concentrations of enterolactone from the three centres were similar, at 6.2, 3.9 and 3.9 ng/ml in samples from Hong Kong Portugal and Britain respectively. **CONCLUSIONS:** Higher concentrations of the isoflavonoid phyto-oestrogens, daidzein and equol, were found in the plasma and prostatic fluid of men from Hong Kong compared to those from Britain and Portugal. However, the levels of the lignan, enterolactone, were very much higher in prostatic fluid of Portuguese men. Isoflavonoids and lignans have many interesting properties and may, in part, be responsible for lower incidences of prostate cancer in men from Asia and also some Mediterranean countries. The isoflavonoids from soya, which are present in high

concentrations in the prostatic fluid of Asian men, may be protective against prostate disease.